Sequential Monte Carlo samplers

Study note for Felipe

26 October 2026

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Prerequisites assumed. Monte Carlo integration with the law of large numbers and the CLT; Bayes' rule, posterior, marginal likelihood; Markov kernels, invariance, detailed balance, Metropolis–Hastings (MH). Everything else is built here.
Scripts. Every number in this note comes from a script in `checks\` (named at each table), with a fixed seed.

1 The problem

1.1 The question

Let E be a space (for example $\mathbb{R}^d$) and let $\pi_1, \pi_2, \ldots, \pi_p$ be probability densities on that *same* space E. Each is known only up to a constant:

$$\pi_n(x) = \frac{\gamma_n(x)}{Z_n}, \qquad Z_n = \int_E \gamma_n(x)\,\mathrm{d}x,$$

where $\gamma_n \geq 0$ can be evaluated at any point x and the normalizing constant Z_n is unknown. The index n is a counter, not necessarily time.

The task is to sample from π_1, then π_2, and so on, and along the way to estimate

(i) expectations $\mathbb{E}_{\pi_n}[\varphi] = \int \varphi(x)\pi_n(x)\,\mathrm{d}x$ for test functions φ, and

(ii) ratios of normalizing constants Z_n/Z_1.

The answer will be a cloud of N weighted points (*particles*) $\{(W_n^i, X_n^i)\}_{i=1}^N$, with $W_n^i > 0$ and $\sum_i W_n^i = 1$, such that $\sum_i W_n^i \varphi(X_n^i) \to \mathbb{E}_{\pi_n}[\varphi]$ as $N \to \infty$.

1.2 Where sequences of targets come from

Del Moral, Doucet and Jasra [10, Section 2.3.1] list four sources.

(a) **Data arriving in order.** $\pi_n(x) = p(x \mid y_1, \ldots, y_n)$, the posterior after n observations.

(b) **A path to one hard target.** The real interest is a single density $\pi = \gamma/Z$ that is hard to sample (several separated modes, high dimension). An easy density μ (normalized, easy to sample) is connected to it by intermediate densities, for example the *geometric path*

$$\gamma_n(x) = \mu(x)^{1-\beta_n}\,\gamma(x)^{\beta_n}, \qquad 0 = \beta_1 < \beta_2 < \cdots < \beta_p = 1. \tag{1}$$

Then $Z_1 = 1$ and $Z_p = Z$, so Z_p/Z_1 is the normalizing constant of the hard target. When γ is prior $\times$ likelihood and μ is the prior, Z is the marginal likelihood. The β_n are called (inverse) *temperatures*, a name borrowed from annealing in physics. The geometric path is a convenience, not a necessity: γ_n can be evaluated wherever μ and γ can, and the ratio $\gamma_n/\gamma_{n-1} = (\gamma/\mu)^{\beta_n - \beta_{n-1}}$ is simple. Other paths exist [7, Section 2.2].

(c) **Optimization.** $\gamma_n = \gamma^{\beta_n}$ with $\beta_n \to \infty$ concentrates on the maxima of γ, as in simulated annealing.

(d) **Rare events.** $\gamma_n(x) = \nu(x)\mathbb{1}\{x \in E_n\}$, for a probability density ν that is easy to sample and nested sets $E = E_1 \supset \cdots \supset E_p = A$; then $Z_p = \nu(A)$, a small probability.

1.3 Why it matters

Case (b) is the central one in statistics. Normalizing constants are how Bayesian models are compared (the Bayes factor is a ratio of two Z's), and multimodal posteriors are where standard samplers fail (Section 2.3). Case (a) is sequential updating without restarting the computation. One method that handles all four cases, and gives Z as a by-product, is what the 2006 paper supplies.

2 Where it came from

2.1 Importance sampling and dimension

The identities. Let q be a density we can sample (the *proposal*) with $q(x) > 0$ wherever $\gamma(x) > 0$ (the *support condition*). Define the *unnormalized weight* $w(x) = \gamma(x)/q(x)$. Then

$$Z = \int \gamma(x)\,\mathrm{d}x = \int \frac{\gamma(x)}{q(x)} q(x)\,\mathrm{d}x = \mathbb{E}_q[w(X)], \qquad \mathbb{E}_\pi[\varphi] = \frac{1}{Z}\int \varphi\gamma = \frac{\mathbb{E}_q[w(X)\varphi(X)]}{\mathbb{E}_q[w(X)]}. \tag{2}$$

The support condition is what makes the middle equality true: a region where $\gamma > 0$ but $q = 0$ is silently dropped from the integral. With $X^1, \ldots, X^N \overset{\text{iid}}{\sim} q$ and $w^i = w(X^i)$,

$$\widehat{Z} = \frac{1}{N}\sum_{i=1}^N w^i, \qquad \widehat{\mathbb{E}_\pi[\varphi]} = \sum_{i=1}^N W^i \varphi(X^i), \quad W^i = \frac{w^i}{\sum_j w^j}. \tag{3}$$

$\widehat{Z}$ is unbiased, $\mathbb{E}[\widehat{Z}] = Z$, by (2). The second estimator (*self-normalized*) is a ratio of two unbiased estimators, $\frac{1}{N}\sum w^i\varphi(X^i)$ over $\frac{1}{N}\sum w^i$. A ratio of unbiased estimators is not unbiased in general, because $\mathbb{E}[A/B] \neq \mathbb{E}[A]/\mathbb{E}[B]$. It is consistent: numerator and denominator converge, so the ratio converges to $\mathbb{E}_\pi[\varphi]$.

How good is it? Write $\bar{w} = w/Z$ for the weight normalized to have $\mathbb{E}_q[\bar{w}] = 1$. Since the w^i are iid,

$$\mathrm{Var}\Big(\frac{\widehat{Z}}{Z}\Big) = \frac{1}{N}\,\mathrm{Var}_q(\bar{w}) = \frac{1}{N}\big(\mathbb{E}_q[\bar{w}^2] - 1\big). \tag{4}$$

So the quality of importance sampling is governed by one number, $\mathbb{E}_q[\bar{w}^2]$. For the self-normalized estimator, the 2006 paper notes that the variance is approximately proportional to $1 + \mathrm{Var}_q(\bar{w}) = \mathbb{E}_q[\bar{w}^2]$ (10, Section 2.1, citing 22, pp. 35–36). Equivalently, it is like having $N/\mathbb{E}_q[\bar{w}^2]$ independent draws from π. This motivates the *effective sample size*

$$\mathrm{ESS} = \frac{\big(\sum_i w^i\big)^2}{\sum_i (w^i)^2} = \frac{1}{\sum_i (W^i)^2} \in [1, N], \tag{5}$$

because $\mathrm{ESS}/N = (\frac{1}{N}\sum w^i)^2/(\frac{1}{N}\sum (w^i)^2)$ estimates $(\mathbb{E}_q w)^2/\mathbb{E}_q[w^2] = 1/\mathbb{E}_q[\bar{w}^2]$. $\mathrm{ESS} = N$ when all weights are equal; $\mathrm{ESS} = 1$ when one particle has all the weight.

The mechanism: mismatch multiplies across coordinates. Suppose target and proposal are products over d coordinates, $\pi(x) = \prod_{j=1}^d \pi_j(x_j)$ and $q(x) = \prod_j q_j(x_j)$. Then $\bar{w}(x) = \prod_j \bar{w}_j(x_j)$ with $\bar{w}_j = \pi_j/q_j$, and because the coordinates are independent under q,

$$\mathbb{E}_q[\bar{w}^2] = \prod_{j=1}^d \mathbb{E}_{q_j}[\bar{w}_j^2] = m^d \quad \text{if every factor equals } m.$$

Each factor satisfies $m = \mathbb{E}[\bar{w}_j^2] \geq (\mathbb{E}[\bar{w}_j])^2 = 1$, with equality only if $\bar{w}_j$ is constant, that is $q_j = \pi_j$. So any fixed mismatch per coordinate makes $\mathbb{E}_q[\bar{w}^2]$, and with it the variance (4), grow *exponentially* in d, and ESS/$N \approx m^{-d}$ shrinks exponentially. This shows that a fixed per-coordinate mismatch is fatal. It does not show that good high-dimensional proposals cannot exist; the problem is that one is rarely available for a hard target.

A computed example. Take $\pi_j = \mathrm{N}(0,1)$ and $q_j = \mathrm{N}(0, s^2)$ with $s = 2$. Then

$$\frac{\pi_j(x)^2}{q_j(x)} = \frac{\frac{1}{2\pi}e^{-x^2}}{\frac{1}{\sqrt{2\pi}\,s}e^{-x^2/(2s^2)}} = \frac{s}{\sqrt{2\pi}}\exp\Big\{-x^2\Big(1-\frac{1}{2s^2}\Big)\Big\},$$

and using $\int e^{-cx^2}\,\mathrm{d}x = \sqrt{\pi/c}$ with $c = 1 - 1/(2s^2)$,

$$m = \int \frac{\pi_j^2}{q_j} = \frac{s}{\sqrt{2\pi}}\sqrt{\frac{\pi}{1-\frac{1}{2s^2}}} = \frac{s}{\sqrt{2-1/s^2}} = \frac{s^2}{\sqrt{2s^2-1}} = \frac{4}{\sqrt{7}} = 1.512.$$

Table 1 uses $N = 10^4$ and 50 repetitions (`s21_is_dimension.py`).

d	m^d	theory ESS/N	simulated ESS/N	theory sd($\widehat{Z}/Z$)	simulated sd($\widehat{Z}/Z$)	mean $\widehat{Z}/Z$
1	1.51	0.66	0.66	0.007	0.007	1.003
5	7.90	0.13	0.13	0.026	0.027	0.997
10	62.4	0.016	0.017	0.078	0.071	0.999
20	3892	2.6×10^{-4}	1.1×10^{-3}	0.62	0.84	1.023
50	9.5×10^{8}	1.1×10^{-9}	2.1×10^{-4}	307	0.068	0.022

Table 1: Importance sampling with a proposal twice as wide as the target in each coordinate.

Up to $d = 10$ theory and simulation agree. At $d = 20$ the empirical ESS already overstates the theoretical one fourfold. At $d = 50$ the simulation *looks* stable (sd 0.068), yet the mean of $\widehat{Z}/Z$ is 0.022 instead of 1. The theoretical variance is carried by enormous weights that occur so rarely that 50 runs never see one; the estimator is unbiased, but its typical value is far too small. The empirical ESS (≈ 2 of 10^4) is the only warning sign. An empirical diagnostic computed from the particles cannot report on regions the particles never reached; this returns in Section 4.

2.2 Moving particles with Markov kernels: the marginal is intractable

A natural repair, and the starting point of the 2006 paper [10, Sections 2.2–2.4], is to build the proposal gradually. Start with weighted particles for an easy π_1. Then move each particle with a Markov kernel $K_n(x, x')$ (the density of the new point x' given the old point x), chosen so that particles drift toward where π_n has mass. If the particles at step $n-1$ have marginal density η_{n-1}, after the move they have marginal density

$$\eta_n(x') = \int \eta_{n-1}(x) K_n(x, x')\,\mathrm{d}x, \qquad \text{so} \qquad \eta_n = \eta_1 K_2 K_3 \cdots K_n. \tag{6}$$

Importance sampling for π_n with proposal η_n needs the weight $\gamma_n(x')/\eta_n(x')$, so it needs $\eta_n(x')$ *pointwise*. That is where it breaks.

The mechanism. Take K_n to be an MH kernel with proposal density $q(x, y)$ and acceptance probability $\alpha(x, y)$. The chain either moves to y (density $q(x, y)\alpha(x, y)$) or stays at x, with probability

$$r(x) = 1 - \int q(x, y)\alpha(x, y)\,\mathrm{d}y.$$

So $K_n(x, \mathrm{d}x') = q(x, x')\alpha(x, x')\,\mathrm{d}x' + r(x)\delta_x(\mathrm{d}x')$, where δ_x is a point mass at x. Plugging into (6),

$$\eta_n(x') = \int \eta_{n-1}(x)\, q(x, x')\alpha(x, x')\,\mathrm{d}x \;+\; \eta_{n-1}(x')\, r(x').$$

Neither term has a closed form. At $n = 2$, η_1 is known, but the first term is still an integral over x with no closed form in general, and the second contains $r(x')$, itself such an integral. From $n = 3$ on, η_{n-1} is itself unknown.

The obvious fix is to replace η_{n-1} by the particles, $\widehat{\eta}_n(x') = \frac{1}{N}\sum_{i=1}^{N} K_n(X_{n-1}^i, x')$. This has two problems. It costs N kernel evaluations per particle, $O(N^2)$ in total. And it still needs $K_n(x, x')$ pointwise, which for MH includes the intractable r [10, Section 2.4]. Kernels whose η_n is tractable exist (independent proposals $K_n(x, x') = K_n(x')$ give $\eta_n = K_n$), but they are rarely good moves on a hard target. The gap is clear: we want local moves such as MCMC, without ever evaluating η_n.

2.3 MCMC: trapped between modes, and no Z

MCMC builds a kernel K that leaves π invariant and runs one long chain. The 2006 paper names its drawbacks for this problem [10, Section 1]: it is hard to tell when the chain has reached its stationary regime, it can stay in a local mode, and it does not by itself handle a sequence of targets.

Separated modes: the mechanism. Take $\pi = \frac{1}{2}\mathrm{N}(-a, 1) + \frac{1}{2}\mathrm{N}(a, 1)$ in one dimension and a random-walk MH chain with proposal sd 2.4. Write $\sigma = 2.4$. To switch modes the chain must cross the low-density valley around 0. It can climb part of the way out of its mode, to $-c$, which costs a factor $e^{-(a-c)^2/2}$ in density. It can then propose a jump to $+c$: a proposal of length $2c$ has density proportional to $e^{-(2c)^2/(2\sigma^2)}$, and it is accepted because the density at $+c$ equals that at $-c$. The cheapest crossing minimizes $f(c) = (a-c)^2/2 + 2c^2/\sigma^2$. Setting $f'(c) = -(a-c) + 4c/\sigma^2 = 0$ gives $c = a\sigma^2/(\sigma^2+4)$, so $a - c = 4a/(\sigma^2+4)$ and

$$f(c) = \frac{8a^2}{(\sigma^2+4)^2} + \frac{2a^2\sigma^2}{(\sigma^2+4)^2} = \frac{2a^2}{\sigma^2+4} = \kappa a^2, \qquad \kappa = \frac{2}{\sigma^2+4} = 0.205.$$

So a crossing has probability of order $e^{-\kappa a^2}$, and the waiting time grows like $e^{0.2a^2}$: exponentially in a^2. Table 2 counts switches in 10^6 steps over 8 chains each (`s23_mcmc_modes.py`).

a	mean wait between switches	log(wait)	$0.205\,a^2$	time in right mode across chains (truth 0.5)
2	9	2.2	0.8	$[0.498, 0.502]$
4	112	4.7	3.3	$[0.491, 0.506]$
6	8,395	9.0	7.4	$[0.442, 0.587]$
7	1.5×10^5	11.9	10.0	$[0.235, 0.679]$
8	4×10^6	15.2	13.1	$[0.000, 0.881]$

Table 2: Random-walk MH on a symmetric two-mode target. The $a = 8$ row rests on one or two observed switches, so its wait is an order of magnitude only.

The waiting time grows by orders of magnitude as the separation grows, and log(wait) tracks κa^2 up to a constant. At $a = 8$ some chains of 10^6 steps never leave their starting mode, and the estimated mode weights range from 0 to 0.88. This is a statement about local moves on this landscape, not a universal law; a cleverer kernel can do better.

No normalizing constant. The MH acceptance ratio $\gamma(y)q(y,x)/\{\gamma(x)q(x,y)\}$ uses only ratios of γ, so Z cancels. That cancellation is why MCMC works without knowing Z, and also why its output carries no direct information about Z. Separate methods exist for estimating Z, such as bridge and path sampling [13].

Sequences of targets. When the target changes from π_{n-1} to π_n, MCMC can restart or continue from where it is, but it has no mechanism to correct for the chain not yet having reached π_n. Weights provide exactly that correction.

2.4 Resampling and particle filters

Importance weights and resampling were developed in a different setting: *state-space models*, where a hidden state x_t evolves and is observed noisily, and the target $p(x_{1:t} \mid y_{1:t})$ lives on a space that grows with t. There, the proposal and the weight are both products over time, so Section 2.1's mechanism applies with t in the role of d, and the weights degenerate as t grows.

Resampling. Rubin's sampling–importance–resampling [25] turns a weighted sample into an unweighted one by sampling indices with probabilities W^i. In SMC one draws N indices $A^1, \dots, A^N$ independently with $\mathbb{P}(A^j = i) = W^i$ and keep $X^{A^1}, \dots, X^{A^N}$ with equal weights $1/N$. Given the current particles, this changes nothing on average:

$$\mathbb{E}\Big[\frac{1}{N}\sum_{j=1}^{N}\varphi(X^{A^j})\,\Big|\,X^{1:N}, W^{1:N}\Big] = \sum_{i=1}^{N} W^i \varphi(X^i).$$

The *bootstrap filter* of Gordon et al. [15] applies this at every time step: propagate each particle through the state dynamics, weight it by the likelihood of the new observation, resample. Dai et al. [7] name Gordon et al. [15] and Kong et al. [20] as the basis of particle filters. The ESS (5) is attributed to Kong et al. [20] by Dai et al. [7] and to Liu and Chen [23] by Del Moral et al. [10, Section 3.1]; neither original was read.

What resampling buys and costs, computed. Local-level model: $x_1 \sim \mathrm{N}(0,1)$, $x_t = x_{t-1} + \mathrm{N}(0,1)$, $y_t = x_t + \mathrm{N}(0,1)$, $t \le 50$, $N = 1000$, 20 data sets. The exact filtering mean $\mathbb{E}[x_t \mid y_{1:t}]$ comes from the Kalman filter (`s24_resampling.py`).

Without resampling, about two particles hold all the weight by $t = 20$ and one by $t = 50$, and the error grows. With resampling, the ESS stays near 630 and the error stays near 0.02: the effort follows the particles that matter. The cost shows in the last column. Resampling duplicates some particles and drops others, so the particles descend from fewer and fewer original ones (15 of 1000 by $t = 50$). Resampling adds variance at the step where it happens and makes the particles dependent. It reallocates effort; it creates nothing new.

	no resampling			resampling every step		
t	ESS	|error|	time-1 ancestors	ESS	|error|	time-1 ancestors
1	648	0.014	1000	656	0.010	530
5	97	0.041	1000	651	0.014	171
10	12	0.202	1000	612	0.011	77
20	2.2	0.500	1000	671	0.022	40
50	1.0	0.847	1000	628	0.022	15

Table 3: Medians over 20 data sets. "Time-1 ancestors" counts the distinct particles at time 1 from which today's particles descend.

2.5 Annealed importance sampling and resample-move

AIS. Neal [24] attacked case (b) directly. Take the path (1) and, for each n, an MCMC kernel K_n that leaves π_n invariant. Draw $X_1 \sim \pi_1$; for $n = 2, \dots, p$, multiply the weight by $\gamma_n(X_{n-1})/\gamma_{n-1}(X_{n-1})$ and then move $X_n \sim K_n(X_{n-1}, \cdot)$. The final weight

$$w = \prod_{n=2}^{p} \frac{\gamma_n(X_{n-1})}{\gamma_{n-1}(X_{n-1})}$$

has an average over independent runs that converges to Z_p/Z_1 [24, Section 2] (in fact $\mathbb{E}[w] = Z_p/Z_1$ exactly, as Section 3.2 shows), so it estimates Z, and the weighted final points estimate $\mathbb{E}_\pi[\varphi]$. The kernel never enters the weight, which is how AIS uses MH moves without the problem of Section 2.2. The same identity appears in nonequilibrium statistical physics as Jarzynski's equality for free-energy differences [17]. Section 3.2 derives this weight as a special case of the 2006 construction.

Where AIS loses: the per-step mismatches multiply. Suppose each move mixed perfectly, so that $X_{n-1} \sim \pi_{n-1}$ independently of the earlier steps. Then the factors of w are independent. Factor n is an importance weight for π_n with proposal π_{n-1}, so by (4) its relative second moment is $1 + v_n$, where

$$v_n = \mathbb{E}_{\pi_{n-1}}\Big[\Big(\frac{\pi_n}{\pi_{n-1}}\Big)^2\Big] - 1 \tag{7}$$

measures how far π_n is from π_{n-1} (it is the χ^2-divergence). Since the factors are independent, $\mathbb{E}[w^2] = \prod_n \mathbb{E}[(\text{factor } n)^2]$ and $\mathbb{E}[w] = \prod_n \mathbb{E}[\text{factor } n]$, so

$$\frac{\mathbb{E}[w^2]}{(\mathbb{E}w)^2} = \prod_{n=2}^{p} (1 + v_n) :$$

the per-step mismatches multiply, as the per-coordinate ones did in Section 2.1. A finer ladder makes every v_n smaller. But each v_n grows with the dimension, real moves do not mix perfectly (the particles lag behind π_n, which adds spread), and nothing in AIS ever resets the weights.

Table 4 runs AIS on

$$\gamma(x) = 4\,\phi_d(x; -4e_1, I) + \phi_d(x; 4e_1, I), \qquad Z = 5,$$

where $\phi_d(\cdot\,; m, I)$ is the d-dimensional standard normal density centred at m (not to be confused with a test function φ) and $e_1 = (1, 0, \dots, 0)$: two unit-variance Gaussian bumps, 8 standard deviations apart along x_1, holding 4/5 and 1/5 of the mass. The start is $\mu = \mathrm{N}(0, 5^2 I)$, with

$d = 1$				$d = 10$		
$p-1$	final ESS/N	mean $\widehat{Z}$	sd $\widehat{Z}$	$p-1$	final ESS/N	sd $\widehat{Z}$
10	0.47	5.01	0.16	10	0.001	47.4
100	0.81	4.99	0.07	100	0.006	3.58
1000	0.97	5.00	0.03	300	0.028	1.58

Table 4: AIS: a finer path helps, but in $d = 10$ the weights stay concentrated on a few particles.

temperatures $\beta_n = ((n-1)/(p-1))^3$ and one random-walk MH step per temperature, $N = 1000$, 100 repetitions (`s25_ais_ladder.py`). The cubic spacing takes small steps near $\beta = 0$, where π_n changes fastest; it is a common heuristic, not a derived choice.

In one dimension AIS works well, and a finer ladder (more intermediate temperatures) helps. In ten dimensions, even with 300 temperatures, the final weights sit on about 28 of 1000 particles: the computation spent on the other particles is nearly wasted.

Resample-move. The step AIS lacks was used in sequential settings by Gilks and Berzuini [14], for dynamic models whose dimension grows, and by Chopin [4], for a static parameter [10, Section 3.5]. When the ESS falls, resample, then apply an MCMC move invariant for the current target, so the duplicates created by resampling spread out again. Chopin's algorithm, iterated batch importance sampling (IBIS), does this for case (a), $\pi_n = p(\theta \mid y_{1:n})$. Left open by these methods: a general account of why these weights are valid, which other moves are allowed, and which weights to use when the move is not an MCMC kernel.

2.6 The 2006 paper and what came after

Del Moral et al. [10] (Pierre Del Moral, Arnaud Doucet, Ajay Jasra; *JRSS B* 68(3), 411–436; received January 2003, final revision January 2006) gave that account. The idea is to do importance sampling not on E but on the space of whole particle paths, using an artificial target built from *backward kernels* (Section 3.1). The intractable marginal of Section 2.2 then never has to be evaluated, any move can be used, and the cost stays $O(N)$. In Section 3.5 of the paper, the weights of AIS, resample-move and IBIS appear as the special case of one particular backward kernel. The paper calls the resulting algorithms *SMC samplers*, and presents them as complementary to MCMC, since MCMC kernels are often their moves [10, Section 1]. It was not the first use of particles on a fixed target: AIS and IBIS came earlier. Dai et al. [7] describe IBIS and SMC samplers as concurrent efforts to generalize particle filters to a much wider class of statistical problems.

Later work that gave it its modern form. Choosing the temperatures automatically from the ESS [18], an idea the 2006 paper already suggests in its conclusion, citing a technical report by Johansen and coauthors that was not read here [10, Section 6]; a conditional ESS criterion [26]; textbook treatment [5]; a review of SMC samplers [7]. The paper notes that its algorithms are particle approximations of a Feynman–Kac flow, for which a general convergence theory exists [10, Section 1], [9].

3 The method, fully explained

3.1 Importance sampling on the path space

The proposal. Run the particles forward as in Section 2.2: $X_1 \sim \eta_1$, then $X_k \sim K_k(X_{k-1}, \cdot)$ for $k = 2, \dots, n$. The marginal η_n of X_n is intractable, but the joint density of the whole path is just a product:

$$\eta_n(x_{1:n}) = \eta_1(x_1) \prod_{k=2}^{n} K_k(x_{k-1}, x_k), \tag{8}$$

where $x_{1:n} = (x_1, \dots, x_n)$. We keep the paper's notation: $\eta_n(x_{1:n})$ is the path density, and $\eta_n(x_n)$ its marginal. So importance sampling on paths is possible, provided we have a target on paths.

The artificial target. We need a density on E^n whose marginal in x_n is π_n, and which can be evaluated up to a constant. Choose, for each k, a *backward kernel* $L_k(x_{k+1}, x_k)$: for each fixed x_{k+1}, a probability density in x_k. Define

$$\widetilde{\pi}_n(x_{1:n}) = \pi_n(x_n) \prod_{k=1}^{n-1} L_k(x_{k+1}, x_k), \qquad \widetilde{\gamma}_n(x_{1:n}) = \gamma_n(x_n) \prod_{k=1}^{n-1} L_k(x_{k+1}, x_k), \tag{9}$$

so $\widetilde{\pi}_n = \widetilde{\gamma}_n / Z_n$ with the same Z_n as before. Read (9) as: draw $x_n \sim \pi_n$, then walk backwards in time, $x_{n-1} \sim L_{n-1}(x_n, \cdot)$, and so on down to x_1.

Its marginal is π_n. Integrate out x_1, then x_2, and so on. Only L_1 contains x_1, and $\int L_1(x_2, x_1)\, \mathrm{d}x_1 = 1$. After that only L_2 contains x_2, and it also integrates to 1. Continuing up to x_{n-1},

$$\int \widetilde{\pi}_n(x_{1:n})\, \mathrm{d}x_{1:n-1} = \pi_n(x_n).$$

So weighted samples of whole paths from $\widetilde{\pi}_n$, with $x_{1:n-1}$ simply discarded, are weighted samples from π_n. And because $\widetilde{\gamma}_n$ integrates to Z_n, the average weight still estimates Z_n. The L_k can be *any* backward kernels; their choice affects the variance only (Section 3.2).

The weights factorize. The importance weight on paths is $w_n(x_{1:n}) = \widetilde{\gamma}_n(x_{1:n}) / \eta_n(x_{1:n})$. Divide it by the weight at the previous step:

$$\begin{aligned} \frac{w_n(x_{1:n})}{w_{n-1}(x_{1:n-1})} &= \frac{\gamma_n(x_n) \prod_{k=1}^{n-1} L_k(x_{k+1}, x_k)}{\eta_1(x_1) \prod_{k=2}^{n} K_k(x_{k-1}, x_k)} \cdot \frac{\eta_1(x_1) \prod_{k=2}^{n-1} K_k(x_{k-1}, x_k)}{\gamma_{n-1}(x_{n-1}) \prod_{k=1}^{n-2} L_k(x_{k+1}, x_k)} \\ &= \frac{\gamma_n(x_n)\, L_{n-1}(x_n, x_{n-1})}{\gamma_{n-1}(x_{n-1})\, K_n(x_{n-1}, x_n)}. \end{aligned}$$

Every factor except the last L and the last K cancels. Define the *incremental weight*

$$\boxed{\widetilde{w}_n(x_{n-1}, x_n) = \frac{\gamma_n(x_n)\, L_{n-1}(x_n, x_{n-1})}{\gamma_{n-1}(x_{n-1})\, K_n(x_{n-1}, x_n)}} \qquad w_n = w_{n-1}\, \widetilde{w}_n. \tag{10}$$

This is the paper's equation (12) [10, Section 3.1]. Each step costs one evaluation per particle, and the total cost is $O(N)$ per step. The marginal η_n appears nowhere. This answers Section 2.2.

The price. We now do importance sampling on E^n instead of E. Weights on a bigger space vary more, unless L is chosen well (next part). For (10) to be a valid importance weight we need $K_n(x_{n-1}, x_n) > 0$ wherever the numerator is positive. A kernel with an atom (MH's rejection) is handled by reading K and L as densities with respect to a common measure that includes the point masses. The choice of L in Section 3.2 sidesteps this, because there K cancels from the ratio.

3.2 Choosing the backward kernel

The optimal choice. Which L makes $\mathrm{Var}(w_n)$ smallest? Under the proposal η_n, decompose by conditioning on the endpoint X_n:

$$\mathrm{Var}(w_n) = \mathbb{E}\big[\,\mathrm{Var}(w_n \mid X_n)\big] + \mathrm{Var}\left(\mathbb{E}[w_n \mid X_n]\right). \tag{11}$$

The conditional density of $x_{1:n-1}$ given x_n under η_n is $\eta_n(x_{1:n})/\eta_n(x_n)$, where $\eta_n(x_n)$ is the marginal. So

$$\mathbb{E}[w_n \mid X_n = x_n] = \int \frac{\widetilde{\gamma}_n(x_{1:n})}{\eta_n(x_{1:n})}\,\frac{\eta_n(x_{1:n})}{\eta_n(x_n)}\,\mathrm{d}x_{1:n-1} = \frac{1}{\eta_n(x_n)}\int \widetilde{\gamma}_n(x_{1:n})\,\mathrm{d}x_{1:n-1} = \frac{\gamma_n(x_n)}{\eta_n(x_n)},$$

using the marginal property just shown. This does not depend on L, so the second term of (11) is fixed. The first term is ≥ 0, and it is 0 exactly when w_n is a function of x_n alone, that is, when $\widetilde{\gamma}_n(x_{1:n})/\eta_n(x_{1:n})$ does not depend on $x_{1:n-1}$. That happens when the backward kernels reproduce the forward chain run backwards.

A Markov chain run backwards is again Markov, with kernels given by Bayes' rule:

$$L_{k-1}^{\mathrm{opt}}(x_k, x_{k-1}) = \frac{\eta_{k-1}(x_{k-1})\,K_k(x_{k-1}, x_k)}{\eta_k(x_k)}. \tag{12}$$

Check: $\eta_n(x_n)\prod_{k=2}^n L_{k-1}^{\mathrm{opt}}(x_k, x_{k-1}) = \eta_n(x_n)\prod_{k=2}^n \frac{\eta_{k-1}(x_{k-1})}{\eta_k(x_k)}K_k(x_{k-1}, x_k)$. The ratios telescope to $\eta_1(x_1)/\eta_n(x_n)$, leaving $\eta_1(x_1)\prod_k K_k = \eta_n(x_{1:n})$. With (12), $\widetilde{\gamma}_n/\eta_n = \gamma_n(x_n)/\eta_n(x_n)$, so $w_n = \gamma_n(x_n)/\eta_n(x_n)$ [10, Proposition 1]. The optimal L brings us back to importance sampling on E with proposal η_n, which is the ideal of Section 2.2 and exactly as intractable, since it contains η. Its use is as a guide: *a good L approximates the forward move run backwards.*

Approximation 1: replace η_{n-1} by π_{n-1}. After resampling, the particles at step $n-1$ are roughly distributed as π_{n-1}. Substituting π_{n-1} for η_{n-1} in (12) gives

$$L_{n-1}(x_n, x_{n-1}) = \frac{\pi_{n-1}(x_{n-1})K_n(x_{n-1}, x_n)}{\int \pi_{n-1}(x)K_n(x, x_n)\,\mathrm{d}x},$$

and in (10) the $K_n(x_{n-1}, x_n)$ cancels and $\pi_{n-1} = \gamma_{n-1}/Z_{n-1}$ gives

$$\widetilde{w}_n = \frac{\gamma_n(x_n)}{\gamma_{n-1}(x_{n-1})} \cdot \frac{\gamma_{n-1}(x_{n-1})/Z_{n-1}}{\int \gamma_{n-1}(x)K_n(x, x_n)\,\mathrm{d}x/Z_{n-1}} = \frac{\gamma_n(x_n)}{\int \gamma_{n-1}(x)K_n(x, x_n)\,\mathrm{d}x} \tag{13}$$

[10, equations (25)–(26)]. This weight depends on where the particle moved to. If K_n mixes perfectly, $K_n(x, x') = \pi_n(x')$, the denominator is $Z_{n-1}\pi_n(x_n)$ and $\widetilde{w}_n = Z_n/Z_{n-1}$, a constant: zero variance. Its drawback is the integral in the denominator. It can be computed only in special cases, such as Gibbs-type updates of one block of coordinates [10, Section 3.3.2.2].

Approximation 2: the reversal of an MCMC kernel. Suppose K_n leaves π_n invariant: $\int \pi_n(x)K_n(x,x')\,\mathrm{d}x = \pi_n(x')$. Put π_n in place of π_{n-1} in Approximation 1. This is a further approximation, good when $\pi_{n-1} \approx \pi_n$ [10, Section 3.3.2.3]. Its payoff is that invariance makes the integral exact: the denominator becomes $\int \pi_n K_n = \pi_n(x_n)$, and

$$L_{n-1}(x_n, x_{n-1}) = \frac{\pi_n(x_{n-1})\, K_n(x_{n-1}, x_n)}{\pi_n(x_n)}. \tag{14}$$

This is a probability density in x_{n-1}, because $\int \pi_n(x_{n-1})K_n(x_{n-1},x_n)\,\mathrm{d}x_{n-1} = \pi_n(x_n)$ by invariance. Only invariance is used, not detailed balance. If K_n does satisfy detailed balance, (14) is $K_n(x_n, x_{n-1})$ itself. Insert (14) into (10):

$$\begin{aligned} \widetilde{w}_n &= \frac{\gamma_n(x_n)}{\gamma_{n-1}(x_{n-1})K_n(x_{n-1},x_n)} \cdot \frac{\pi_n(x_{n-1})K_n(x_{n-1},x_n)}{\pi_n(x_n)} = \frac{\gamma_n(x_n)}{\pi_n(x_n)} \cdot \frac{\pi_n(x_{n-1})}{\gamma_{n-1}(x_{n-1})} \\ &= Z_n \cdot \frac{\gamma_n(x_{n-1})}{Z_n\,\gamma_{n-1}(x_{n-1})} = \boxed{\frac{\gamma_n(x_{n-1})}{\gamma_{n-1}(x_{n-1})}} \end{aligned} \tag{15}$$

[10, equations (30)–(31)]. Three consequences:

(i) The kernel cancels completely. Any MCMC kernel can be used, MH with its rejection atom included, without evaluating its density.

(ii) This is the AIS weight of Section 2.5. So $\mathbb{E}[w] = Z_p/Z_1$ for AIS follows from Section 3.1: AIS is importance sampling on paths with this L [10, Section 3.5]. Resample-move and IBIS use the same weight.

(iii) The weight depends on the old point only, so it can be computed *before* the move. Then one can resample first and move afterwards, and the move rejuvenates the duplicates [10, Remark 1].

What it gives up. Because (15) ignores the move, even a perfect kernel cannot reduce its variance: if $K_n = \pi_n$ exactly, the weights $\gamma_n(X_{n-1})/\gamma_{n-1}(X_{n-1})$ still vary as much as π_n/π_{n-1} does, whereas (13) would have been constant [10, Section 3.5]. So with (15) successive targets must be close, which means many intermediate distributions. In practice (15) is the standard choice anyway, because it is the one that is always computable.

3.3 Degeneracy, ESS and resampling

Weights multiply along the path by (10), so the spread mechanism of Sections 2.1 and 2.4 applies, with the number of steps in the role of the dimension. The remedy is the one of Section 2.4: monitor ESS (5), and when it falls below a threshold (commonly $N/2$), resample and reset all weights to $1/N$ [10, Section 3.1].

Schemes. Any rule that gives particle i a random number N^i of copies, with $\sum_i N^i = N$ and $\mathbb{E}[N^i \mid W] = NW^i$, keeps the conditional unbiasedness of Section 2.4. Let $C_i = W^1 + \cdots + W^i$ ($C_0 = 0$).

- **Multinomial:** N independent uniforms U^j; copy particle i for each $U^j \in (C_{i-1}, C_i]$.

- **Systematic:** one uniform $U \sim \mathrm{Unif}(0,1)$ and the N evenly spaced points $u_j = (U+j-1)/N$, $j = 1, \ldots, N$; copy particle i for each $u_j \in (C_{i-1}, C_i]$. Each u_j is uniform on $((j-1)/N,\ j/N]$, so $\mathbb{P}(u_j \in (C_{i-1}, C_i]) = N\times$(length of the overlap). Summing over j gives $\mathbb{E}[N^i] = NW^i$. Because the points are $1/N$ apart, an interval of length W^i contains either $\lfloor NW^i \rfloor$ or $\lceil NW^i \rceil$ of them, so each count varies by at most one. This does not guarantee a smaller variance than multinomial: the counts are dependent, and Douc et al. [11] give an example where systematic resampling is worse. It is popular because it is simple and fast. Douc et al. [11] credit this single-uniform scheme to Carpenter et al. [3], who called it "stratified".
- **Residual** and **stratified** resampling are shown by Douc et al. [11] to improve on multinomial; for stratified resampling they refer to Kitagawa [19].

Resampling always adds some noise, so resampling when the weights are nearly equal only adds variance. That is why it is triggered by ESS rather than done at every step; the paper notes that neither of its two asymptotic variances (Section 3.6) is smaller in general [10, Remark 4].

3.4 The algorithm and the normalizing constant

SMC sampler (general form; 10, Section 3.1.1)

1. *Initialize.* Draw $X_1^i \sim \eta_1$, set $w_1^i = \gamma_1(X_1^i)/\eta_1(X_1^i)$ and normalize to W_1^i, for $i = 1, \ldots, N$.
2. For $n = 2, \ldots, p$:
   (a) *Resample* if ESS $< \tau N$, for a threshold $\tau \in (0,1)$ (commonly $\frac{1}{2}$); set $W_{n-1}^i = 1/N$.
   (b) *Move:* $X_n^i \sim K_n(X_{n-1}^i, \cdot)$.
   (c) *Reweight:* $\widetilde{w}_n^i = \widetilde{w}_n(X_{n-1}^i, X_n^i)$ by (10); record $\widehat{r}_n = \sum_i W_{n-1}^i \widetilde{w}_n^i$; set $W_n^i = W_{n-1}^i \widetilde{w}_n^i / \widehat{r}_n$.
3. *Output.* $\pi_n \approx \sum_i W_n^i \delta_{X_n^i}$, and $\widehat{Z_n/Z_1} = \prod_{k=2}^n \widehat{r}_k$ when $\eta_1 = \pi_1$.

Why $\widehat{r}_n$ estimates Z_n/Z_{n-1}. The weighted paths $(W_{n-1}^i, X_{1:n-1}^i)$ target $\widetilde{\pi}_{n-1}$; after the move, the extended paths target $\widetilde{\pi}_{n-1}(x_{1:n-1})K_n(x_{n-1}, x_n)$. Under that density, the expected incremental weight is

$$\begin{aligned}
&\int \widetilde{\pi}_{n-1}(x_{1:n-1})K_n(x_{n-1}, x_n)\, \frac{\gamma_n(x_n)L_{n-1}(x_n, x_{n-1})}{\gamma_{n-1}(x_{n-1})K_n(x_{n-1}, x_n)}\, \mathrm{d}x_{1:n} \\
&\quad = \int \pi_{n-1}(x_{n-1})\, \frac{\gamma_n(x_n)L_{n-1}(x_n, x_{n-1})}{\gamma_{n-1}(x_{n-1})}\, \mathrm{d}x_{n-1}\, \mathrm{d}x_n \\
&\quad = \frac{1}{Z_{n-1}} \int \gamma_n(x_n) \Big[\int L_{n-1}(x_n, x_{n-1})\, \mathrm{d}x_{n-1} \Big]\, \mathrm{d}x_n = \frac{Z_n}{Z_{n-1}}.
\end{aligned}$$

In the first line, $x_{1:n-2}$ is integrated out using the marginal property of $\widetilde{\pi}_{n-1}$, and the K_n's cancel. In the second, $\pi_{n-1}/\gamma_{n-1} = 1/Z_{n-1}$ and L_{n-1} integrates to 1. $\widehat{r}_n$ is the weighted particle average of this expectation [10, equation (14)], and the product of the ratios telescopes to Z_n/Z_1.

The common special case. With the MCMC-reversal weight (15), the weight needs only X_{n-1}, so the steps run in the order **reweight** → **resample if needed** → **move**:

$$\widetilde{w}_n^i = \frac{\gamma_n(X_{n-1}^i)}{\gamma_{n-1}(X_{n-1}^i)}, \quad \widehat{r}_n = \sum_i W_{n-1}^i \widetilde{w}_n^i, \quad W_n^i \propto W_{n-1}^i \widetilde{w}_n^i;$$

then resample if ESS $< \tau N$; then move each particle with K_n. This order belongs to the invariant-kernel case; with a general L, the move must come before the weight, as in the box.

Unbiasedness of $\widehat{Z}$. The estimator $\prod_k \widehat{r}_k$ of Z_n/Z_1 is *unbiased* under the following conditions: the targets and the kernels are fixed in advance (not computed from the particles); $\eta_1 = \pi_1$ is sampled exactly; and resampling satisfies $\mathbb{E}[N^i \mid W] = NW^i$. Dai et al. [7] state the unbiasedness, and note that resampling triggered by the ESS (rather than at fixed times) on a fixed path does not alter it; Dau and Chopin [8, Proposition 2] trace the result to Del Moral's Feynman–Kac theory. We do not prove it here; Section 3.7(a) checks it numerically. Two estimators that are *not* unbiased:

- The self-normalized expectations $\sum_i W_n^i \varphi(X_n^i)$, for the reason given in Section 2.1.
- $\log \widehat{Z}$. By Jensen's inequality, $\mathbb{E}[\log \widehat{Z}] \le \log \mathbb{E}[\widehat{Z}] = \log Z$, so it is biased downwards.

When the temperatures or kernels are chosen from the particles themselves (next part), the fixed-in-advance condition fails and this result no longer applies (Section 4.5).

3.5 Tempered SMC for one fixed target

The most common use is case (b): the geometric path (1), MCMC moves, and weights (15). Write $\ell(x) = \log \gamma(x) - \log \mu(x)$, as in Section 2.5. Then $\gamma_n(x)/\gamma_{n-1}(x) = \exp\{(\beta_n - \beta_{n-1})\,\ell(x)\}$, so

$$\widetilde{w}_n^i = \exp\{\delta_n\, \ell(X_{n-1}^i)\}, \qquad \delta_n = \beta_n - \beta_{n-1}.$$

When μ is the prior and γ is prior × likelihood, ℓ is the log-likelihood and γ_n is the *power posterior* (prior × likelihood$^{\beta_n}$). Figure 1 shows the path for the two-bump target of Section 2.5. Early on, π_n is wide and unimodal and a random walk moves freely. The two bumps emerge gradually, and the particles are already sitting in both by then.

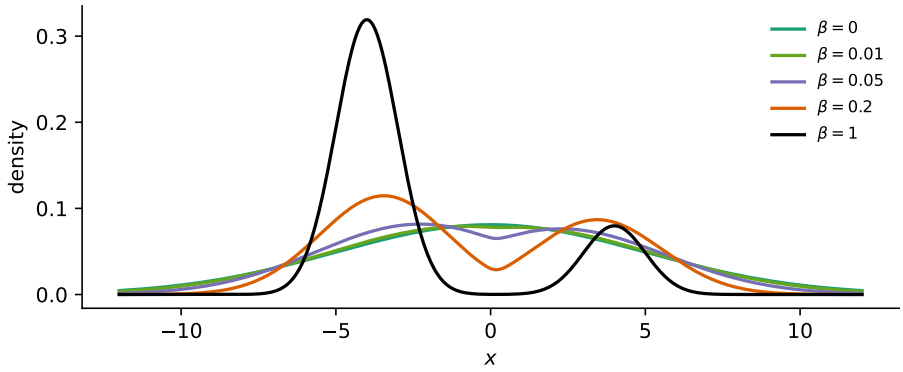


Figure 1: The geometric path from $\mu = \mathrm{N}(0, 5^2)$ to the two-bump target of Section 2.5 in $d = 1$, $\pi \propto 4\phi_1(x; -4, 1) + \phi_1(x; 4, 1)$ (`fig_path.py`).

Choosing the temperatures adaptively. How large should each step δ_n be? Too large, and the weights degenerate; too small, and computation is wasted. The ESS answers this on the fly. Given particles X^i at β_{n-1} with weights W^i, the ESS that a step δ would produce is

$$\mathrm{ESS}(\delta) = \frac{\left(\sum_i W^i e^{\delta \ell_i}\right)^2}{\sum_i (W^i)^2 e^{2\delta \ell_i}}, \qquad \ell_i = \ell(X^i).$$

With resampling at every step (so $W^i = 1/N$), choose δ so that $\mathrm{ESS}(\delta) = \alpha N$ for a target fraction $\alpha \in (0,1)$ (for example $\frac{1}{2}$), or $\delta = 1 - \beta_{n-1}$ if even the full remaining step keeps ESS $\geq \alpha N$ (18; in this form, 7, Section 2.4). When the incoming weights W^i are unequal, $\mathrm{ESS}(\delta)$ mixes the old inequality with the new step's. In particular $\mathrm{ESS}(0)$ is the incoming ESS, which may already be below αN, and then no step solves the equation. Zhou et al. [26] proposed the *conditional* ESS, which measures the new step alone:

$$\mathrm{CESS}(\delta) = \frac{N\left(\sum_i W^i e^{\delta \ell_i}\right)^2}{\sum_i W^i e^{2\delta \ell_i}}.$$

With equal incoming weights $W^i = 1/N$, both reduce to $(\sum_i e^{\delta \ell_i})^2 / \sum_i e^{2\delta \ell_i}$.

Bisection works because the criterion decreases in δ. Let p_s be the distribution on $\{1, \ldots, N\}$ with $p_s(i) \propto W^i e^{s\ell_i}$, and let $M(s) = \log \sum_i W^i e^{s\ell_i}$. Differentiating, $M'(s) = \mathbb{E}_{p_s}[\ell]$ and $M''(s) = \mathrm{Var}_{p_s}(\ell) \geq 0$, so M' is nondecreasing. Then

$$\frac{\mathrm{d}}{\mathrm{d}\delta} \log \mathrm{CESS}(\delta) = 2M'(\delta) - 2M'(2\delta) \leq 0 \quad \text{for } \delta \geq 0,$$

since $2\delta \geq \delta$. So CESS is nonincreasing in δ, and with equal weights so is ESS. Hence $\{\delta : \mathrm{CESS}(\delta) \geq \alpha N\}$ is an interval starting at 0, and bisection finds its right end, as in Dai et al. [7, Section 2.4]; Zhou et al. [26] use the same binary search on the CESS.

Choosing the moves. The kernel K_n is any MCMC kernel invariant for π_n: random-walk MH, Gibbs, or one of the many others. Its parameters can be set from the current particles, for example a random-walk covariance equal to the particle covariance [7, Section 2.3], citing Chopin [4]. Having an approximation of the current target in hand is a practical advantage over MCMC. As with the temperatures, choosing it from the particles breaks the fixed-in-advance condition of Section 3.4.

3.6 What the theory says about the error

The CLT. Under integrability conditions, $\sqrt{N}\left(\sum_i W_n^i \varphi(X_n^i) - \mathbb{E}_{\pi_n}[\varphi]\right)$ and $\sqrt{N}\left(\log \widehat{Z_n/Z_1} - \log(Z_n/Z_1)\right)$ converge to centred normal distributions [10, Proposition 2]. The paper gives the asymptotic variances in two extreme cases. Without resampling, the variance is that of importance sampling on the path space. With multinomial resampling at every step, it is a sum of one term per step. For an annealing sequence under mixing assumptions, the paper notes that the first can grow without bound in n while the second stays bounded [10, Remark 4]. The particles are dependent (resampling couples them), so these are not the iid CLT. The variances are not simple functions of the particle cloud, which is why estimating them from one run is hard (Section 4.4).

The idealized case, computed. To see what drives the error in $\widehat{Z}$, take the special case of Section 3.4, resample at every step, and assume perfect mixing, $K_{n-1}(x, x') = \pi_{n-1}(x')$ (for $n = 2$, $X_1 \sim \pi_1$ exactly). Then, before reweighting at step n, the particles are exactly iid from π_{n-1}, independent of the past. So $\widehat{r}_n = \frac{1}{N}\sum_i \gamma_n(X^i)/\gamma_{n-1}(X^i)$ is an importance sampling estimate of Z_n/Z_{n-1} with proposal π_{n-1}, and the $\widehat{r}_n$ are independent across n. By (4),

$$\frac{\mathrm{Var}(\widehat{r}_n)}{(Z_n/Z_{n-1})^2} = \frac{v_n}{N},$$

with v_n the divergence (7) of Section 2.5. For a product of independent factors, $\mathbb{E}[\prod \widehat{r}_n^2] = \prod \mathbb{E}[\widehat{r}_n^2]$. So the relative variance of $\widehat{Z} = \prod_n \widehat{r}_n$ is

$$\frac{\mathrm{Var}(\widehat{Z})}{Z^2} = \prod_{n=2}^{p}\Big(1 + \frac{v_n}{N}\Big) - 1 \;\approx\; \frac{1}{N}\sum_{n=2}^{p} v_n \qquad \Big(\text{when } \sum_n v_n \ll N\Big). \tag{16}$$

Compare AIS under the same idealization (Section 2.5): the average of N independent AIS weights has relative variance $\{\prod_n(1+v_n) - 1\}/N \approx (e^{\sum_n v_n} - 1)/N$. *Resampling turns the product of the per-step mismatches into a sum.* This is what resampling buys, and it answers the failure of AIS shown in Section 2.5. Even with perfect moves, the error in $\widehat{Z}$ is set by how far apart successive targets are, summed along the path. The paper makes the same point from its variance formulas [10, Section 4.2.3]. Two consequences: the path must be fine enough, and adaptive temperature choice, which holds each step's ESS $\approx N/(1+v_n)$ at αN, is in effect fixing each v_n at about $1/\alpha - 1$ [7, Section 2.4].

3.7 Worked examples

(a) By hand, on three points. Let $E = \{1, 2, 3\}$ and $\gamma_n(x) = (x^2)^{\beta_n}$ with $\beta_1 = 0$, $\beta_2 = \frac{1}{2}$, $\beta_3 = 1$:

$$\gamma_1 = (1, 1, 1), \quad \gamma_2 = (1, 2, 3), \quad \gamma_3 = (1, 4, 9); \qquad Z_1 = 3,\ Z_2 = 6,\ Z_3 = 14.$$

The target ratio is $Z_3/Z_1 = 14/3 = 4.667$. Use $N = 3$ particles, the weight (15), systematic resampling at every step, and one MH move invariant for the current target, which proposes one of the other two states uniformly (`s37a_finite.py`).

1. *Start.* Draw from π_1 (uniform): $X = (1, 1, 3)$, weights $1/3$ each.

2. *Reweight to $\beta_2 = \frac{1}{2}$.* $\widetilde{w}^i = \gamma_2(X^i)/\gamma_1(X^i) = (1, 1, 3)$. Then $\widehat{r}_2 = \frac{1}{3}(1 + 1 + 3) = \frac{5}{3}$ (truth $Z_2/Z_1 = 2$). The normalized weights are $W = (\frac{1}{5}, \frac{1}{5}, \frac{3}{5})$ and ESS $= 1/(\frac{1}{25} + \frac{1}{25} + \frac{9}{25}) = \frac{25}{11} = 2.27$.

3. *Resample (systematic, $U = 0.5$).* Points $u_j = (0.5 + j - 1)/3 = (0.167, 0.5, 0.833)$; cumulative weights $C = (0.2, 0.4, 1)$. $0.167 \in (0, 0.2] \to$ particle 1; 0.5 and $0.833 \in (0.4, 1] \to$ particle 3. New $X = (1, 3, 3)$, weights $1/3$: the particle at 1 with weight $\frac{1}{5}$ survives once, the one at 3 with weight $\frac{3}{5}$ is copied twice ($NW = 1.8$, so $\lceil 1.8 \rceil = 2$ copies).

4. *Move (MH for $\gamma_2 = (1, 2, 3)$).* Particle 1 at 1 proposes 2: acceptance $\min(1, 2/1) = 1$, move to 2. Particle 2 at 3 proposes 1: acceptance $\frac{1}{3}$; with uniform 0.7, reject, stay at 3. Particle 3 at 3 proposes 2: acceptance $\frac{2}{3}$; with uniform 0.4, accept, move to 2. Now $X = (2, 3, 2)$: the two copies have separated.

5. *Reweight to $\beta_3 = 1$.* $\widetilde{w}^i = \gamma_3(X^i)/\gamma_2(X^i) = X^i = (2, 3, 2)$, so $\widehat{r}_3 = \frac{7}{3}$ (truth $Z_3/Z_2 = 14/6 = \frac{7}{3}$). The weights are $W = (\frac{2}{7}, \frac{3}{7}, \frac{2}{7})$.

The estimate is $\widehat{Z_3/Z_1} = \widehat{r}_2\,\widehat{r}_3 = \frac{5}{3}\cdot\frac{7}{3} = \frac{35}{9} = 3.89$ (truth 4.667). The mean under π_3 is estimated as $\widehat{\mathbb{E}_{\pi_3}X} = \frac{2\cdot2+3\cdot3+2\cdot2}{7} = \frac{17}{7} = 2.43$ (truth $\frac{1+8+27}{14} = \frac{18}{7} = 2.57$). One run with three particles is noisy. Repeating the whole algorithm 10^6 times gives a mean of $\widehat{Z}$ equal to 4.6660 (standard error 0.0014), matching $14/3 = 4.6667$: unbiased, as stated in Section 3.4. The mean of $\log\widehat{Z}$ is 1.495, below $\log(14/3) = 1.540$: biased downwards, as Jensen's inequality says.

(b) Two Gaussian bumps in $d = 2$ and $d = 10$. The target is

$$\gamma(x) = 4\,\phi_d(x; -4e_1, I) + \phi_d(x; 4e_1, I), \qquad Z = 5\ (\log Z = 1.609),$$

the target of Section 2.5. The reported quantity is $\mathbb{P}_\pi(x_1 > 0) = (4\Phi(-4) + \Phi(4))/5 = 0.20002$, essentially the weight of the minor bump. Every method starts from $\mu = \mathrm{N}(0, 5^2 I)$. Random-walk MH proposals have scale $2.38/\sqrt{d}$ (the standard random-walk scaling heuristic) times the standard deviation of one tempered bump, $((1-\beta)/25 + \beta)^{-1/2}$. This rule is fixed in advance, so the fixed-ladder runs satisfy the conditions of Section 3.4. Each method gets the same budget of target evaluations: $N = 500$ particles, $S = 5$ MH steps per temperature, and a fixed ladder $\beta_n = ((n-1)/(p-1))^3$ with $p - 1 = 20$ steps ($d = 2$) or 100 ($d = 10$). That gives about 5×10^4 and 2.5×10^5 evaluations. IS uses that many draws from μ, and MH runs one chain of that length (started from a draw of μ, scale $2.38/\sqrt{d}$, all steps kept). AIS and SMC share the ladder and kernels, so *resampling is the only difference* between them; SMC uses systematic resampling when ESS $< N/2$. The adaptive sampler uses $\alpha = 1/2$ and systematic resampling at every step. With $S = 5$ it takes few steps and uses far fewer evaluations than the others, so it is also run with $S = 65$, chosen so that its budget matches. There are 100 independent runs of each (`s37b_mixture.py`).

$d = 10$	$\mathbb{P}(x_1 > 0)$, truth 0.200		$\widehat{Z}$, truth 5			
method	mean	sd	mean	sd	sd $\log\widehat{Z}$	notes
IS from μ	0.264	0.206	4.62	3.31	0.69	ESS a few, of 250,500
one MH chain	0.228	0.124	–	–	–	7% of chains stuck
AIS	0.200	0.060	5.03	1.15	0.22	final ESS $\approx$ 37 of 500
SMC, same ladder	0.198	0.027	5.02	0.49	0.10	3 resamplings; 173 ancestors
adaptive SMC, $S = 5$	0.216	0.139	4.71	3.73	0.60	7.7 steps; 19,700 evals
adaptive SMC, $S = 65$	0.197	0.025	4.98	0.62	0.13	7.9 steps; 256,600 evals

Table 5: The two-bump target in $d = 10$, 100 runs per method. "Stuck": estimate < 0.05 or > 0.95. "Ancestors": distinct initial particles that the final particles descend from.

Reading.

- **IS** has collapsed: Section 2.1's mechanism in ten dimensions.
- **The single MH chain** rarely crosses between bumps: Section 2.3's mechanism.
- **AIS** is roughly unbiased for Z but noisy, with its weight on about 37 particles.
- **SMC on the same ladder** halves the error in both $\widehat{Z}$ and $\mathbb{P}(x_1 > 0)$ (sd 0.49 vs 1.15, and 0.027 vs 0.060). The only change is about three resampling steps, which move effort away from particles with negligible weight. The 2006 paper's own mixture example found that resampling improved posterior expectations but not $\widehat{Z}$ [10, Section 4.2.4]. How much resampling helps $\widehat{Z}$ depends on the problem.

- **The adaptive sampler** takes only about 8 large steps. With $S = 5$ moves per step it is poor: after each resampling, five random-walk steps cannot spread the duplicates out again (on average only 25 distinct original particles survive). Given the same budget ($S = 65$), it matches fixed-ladder SMC for $\mathbb{P}(x_1 > 0)$ and comes close for $\widehat{Z}$: fewer, larger steps mean larger v_n (Section 3.6). Adaptive temperatures decide *where* to step, not how well the particles are rejuvenated.

In $d = 2$ the same experiment (`out_s37b_d2.txt`) shows the other side: plain IS is the most accurate method there (sd of $\mathbb{P}(x_1 > 0)$ 0.004 against 0.021–0.049 for the others). When the dimension is low and the start distribution covers the target, sequential machinery is wasted effort.

4 Drawbacks

4.1 Particles can only be reweighted where they already are

Weights rescale existing particles; they cannot put a particle where none went. Rerun Section 3.7(b) with the start distribution centred on the *major* bump, $\mu = \mathrm{N}(-4e_1, I)$. Then $\gamma/\mu \approx 4$ wherever particles are, so every weight is nearly the same. The adaptive sampler sees ESS $\approx N$ and jumps straight to $\beta = 1$. Over 100 runs in $d = 10$ (random-walk scale $2.38/\sqrt{d}$): the mean of $\widehat{Z}$ is 4.000 with sd under 0.01, and $\mathbb{P}(x_1 > 0) = 0.0002$, against truths of 5 and 0.200. Every diagnostic says the run went perfectly.

The mechanism is Section 2.1's. Here the sampler took a single step, so its $\widehat{Z}$ is in effect plain importance sampling from μ, which is unbiased because $\mu > 0$ everywhere. But the missing mass of 1 sits on particles near the minor bump, whose weight is about $\gamma(4e_1)/\mu(4e_1) = \phi_1(0)/\phi_1(8) = e^{32}$, and the density of μ there is of order e^{-32}, so such particles essentially never appear. ESS measures the spread among weights that were drawn, so it cannot see this. The protection is a start distribution and a path that cover all the mass (here, μ wide enough), plus moves able to cross between regions. Comparing several independent runs from different start distributions also helps.

4.2 The moves carry the exploration

Resampling only duplicates particles. Diversity comes back only through K_n, so SMC is no better than its moves allow. The adaptive rows of Table 5 show this: the same adaptive rule with 5 or with 65 MH steps per temperature gives sd 0.139 or 0.025. Near $\beta = 1$ the kernel faces the full difficulty of the final target. SMC does not need the kernel to cross between modes (the tempering moved particles into both bumps while π_n was still wide), but it does need the kernel to rejuvenate within them. If it cannot, the particles collapse onto a few ancestors.

4.3 Cost

The cost is about $N \times (p-1) \times S$ target evaluations (S MH steps per temperature): 2.5×10^5 in Table 5, the length of an MCMC chain that, on an easier target, might have done the job alone. The work within a step is parallel across particles. Resampling needs all particles at once, so it is a synchronization point.

4.4 The error is hard to assess from one run

Resampling makes the particles dependent, so iid standard-error formulas are wrong, and the CLT variances of Section 3.6 are not simple functions of one particle cloud. The reliable route is the

one used throughout this note: repeat the whole algorithm independently and look at the spread. That multiplies the cost. Single-run variance estimators exist (Section 6).

4.5 Adaptation and exactness

Choosing temperatures or move scales from the particles breaks the fixed-in-advance condition of Section 3.4: the estimate of Z "need not be unbiased when adaptive techniques are employed" [26, Section 3.3.2]. Adapting the kernels does not affect consistency as $N \to \infty$, but may not preserve unbiasedness [7, Section 2.3]. Where exact unbiasedness matters, the remedy follows from the condition itself and is the one Dai et al. [7] give. Run once adaptively, record the ladder and kernel settings, then run again with fresh random numbers and those settings frozen, at about twice the cost. In the second run, everything is fixed in advance relative to its own randomness.

4.6 Dimension

Equation (16) shows how dimension enters. For a small step δ on the geometric path, $\pi_n/\pi_{n-1} \propto e^{\delta\ell}$, and expanding to second order in δ gives $v_n \approx \delta^2 \operatorname{Var}_{\pi_{n-1}}(\ell)$. If ℓ is a sum over d roughly independent coordinates, $\operatorname{Var}(\ell) \le c\,d$ for some constant c. With $p-1$ equal steps, $\delta = 1/(p-1)$, so

$$\sum_n v_n \approx (p-1) \cdot \frac{1}{(p-1)^2} \cdot c\,d = \frac{c\,d}{p-1}.$$

Keeping the error fixed therefore needs $p \propto d$ intermediate distributions. The number of target evaluations then grows linearly in d (each evaluation costs $O(d)$, so the arithmetic is $O(Nd^2)$), rather than exponentially as for importance sampling in Section 2.1. The expansion needs $\delta^2 \operatorname{Var}(\ell)$ small, which holds when $\delta \sim 1/d$. The argument inherits the perfect-mixing idealization of (16); real moves must also mix in d dimensions, which usually takes more MCMC steps as d grows. Beskos et al. [2] prove a result of this kind. With $O(d)$ tempering steps and cost $O(Nd^2)$, the ESS stays stable as $d \to \infty$, and fewer steps make it collapse. Their assumptions are a product target of d iid coordinates and kernels acting coordinate by coordinate. It is a scaling under conditions, not a universal law.

4.7 Many choices

Performance depends heavily on the sequence $\{\pi_n\}$, the forward kernels $\{K_n\}$ and the backward kernels $\{L_n\}$ [10, Section 6], plus N, α, the resampling threshold τ and the number of MH steps S. Adaptive rules cover the temperatures and scales; the rest is judgement.

5 When to use it, and when not

5.1 Use an SMC sampler when

- **The target is hard, and an easy distribution connects to it.** Examples are several modes, or a posterior far from the prior. The tempering path lets particles settle into all regions while the target is still smooth (Figure 1, Table 5).
- **You need the normalizing constant**, for example a marginal likelihood for model comparison. It comes out of the same run (Section 3.4).
- **The targets come in a sequence**, such as posteriors as data batches arrive. The previous particles are reused instead of restarting.

- **You have parallel hardware.** The work within a step is parallel across particles.
- **You want automatic tuning.** Temperatures and move scales can be set from the particles (Section 3.5).

5.2 Use something else when

- **One target, a well-mixing MCMC, and no need for Z.** One chain is cheaper.
- **Low dimension and a proposal that covers the target.** Plain importance sampling wins (the $d = 2$ case of Section 3.7).
- **γ is very expensive.** Then $N \times p \times S$ evaluations are out of reach. Methods that need fewer evaluations (MCMC, or approximations such as Laplace or variational methods) may be the only option.
- **The unknowns are hidden states evolving in time** (state-space models). Particle filters are SMC specialized to that structure (Section 6).
- **No good path exists.** If no start distribution and path reach all modes, SMC will miss them silently (Section 4.1). Use moves designed for the specific structure, and compare runs from several starts.

6 What next

Three branches. The first continues this note directly.

6.1 Practical SMC samplers

- **Variance from a single run.** Estimates the variance of SMC outputs from the particle genealogy, without repeated runs. Lee and Whiteley [21].
- **Waste-free SMC.** Keeps the whole MCMC trajectory of a few resampled particles instead of only the endpoints, which uses the move effort better and makes performance less sensitive to the number of MCMC steps. Dau and Chopin [8].

6.2 Latent-state models

- **Particle filtering and smoothing.** SMC for state-space models, the setting of Section 2.4. Gordon et al. [15]; tutorial by Doucet and Johansen [12].
- **Particle MCMC.** Uses an SMC estimate of the likelihood inside MCMC for parameters, exactly, thanks to unbiasedness. Andrieu et al. [1].
- **SMC^2.** An SMC sampler over parameters whose weights come from particle filters. Chopin et al. [6].

6.3 Theory and design

- **Feynman–Kac theory.** The general convergence theory for particle approximations, to which the 2006 paper refers its results [10, Section 1]. Del Moral [9].
- **Stability in high dimensions.** When and how fast tempered SMC scales with d. Beskos et al. [2].
- **Controlled SMC.** Chooses the proposals by approximately solving an associated optimal control problem with an iterative scheme, improving on fixed proposals at the same cost. Heng et al. [16].

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